

Fair & Unbiased Assessment System

Database Design Document

V 1.3

Submitted By

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REVISION HISTORY

Date	Version	Description	Approved by
15/05/26	V 1.3	Finalized the report by updating relationship names, cardinalities and PK identification, references, and AI appendix section. Expanded entity attribute documentation from basic entities to all final entities of the ERD and removed associative entities from the entity list table	Group Members
12/05/26	V 1.2	Updated the report according to the modified ERD structure by introducing Program, Batch, Session, and Section entities. Improved normalization, modified Enrollment structure, and updated business rules and assumptions according to university workflow.	Ma'am Asiya
10/05/26	V 1.1	Prepared the initial ERD report including introduction, objectives, entity list and attributes. Initial version also included associative entities in the entity table and limited attribute documentation.	Self

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CHAPTER 1: PROJECT OVERVIEW

1.1. INTRODUCTION:

The purpose of this milestone is to design and develop a complete Entity Relationship Diagram (ERD/EERD) for the proposed AI-Based Academic Assessment and Report Generation System. This phase focuses on identifying the major entities, defining their attributes, and establishing relationships to represent the complete database structure using standard ERD/EERD notation.

The proposed system is designed to manage academic activities within a university environment, including student enrollment, faculty teaching assignments, course management, assessments, submissions, evaluations, and AI-generated academic reports. The ERD acts as the foundational blueprint of the system by organizing how academic data is stored, connected, and maintained throughout the workflow.

During the design process, special attention was given to real-world university scenarios such as program and batch management, semester and session tracking, section handling, repeat-course enrollment, and preservation of historical academic records. Associative entities such as Enrollment, Teacher Assignment, and Registered Courses were introduced to support accurate academic tracking and maintain proper many-to-many relationships. This milestone provides the structural foundation required for implementing the database system in later phases of the project.

1.2. PROBLEM STATEMENT:

Traditional academic management and assessment systems used in many educational institutions mainly focus on storing student records, course information, and examination data while lacking intelligent mechanisms for structured assessment management and automated academic analysis. In most cases, assessment evaluation, feedback generation, and performance analysis are performed manually by instructors, making the process time-consuming, repetitive, and difficult to manage efficiently for large classes.

Additionally, many systems fail to properly maintain academic relationships such as program structures, teaching assignments, enrollment history, batch management, semester progression, and repeat-course scenarios. Poor database design can lead to data redundancy, inconsistency, weak scalability, and difficulties in preserving historical academic records.

The proposed AI-Based Academic Assessment and Report Generation System require a well-structured and normalized database design capable of accurately representing complex university workflows and relationships. Therefore, there is a need to design a comprehensive Entity Relationship Diagram (ERD/EERD) that identifies all major entities, their attributes, associative entities, subtype-supertype structures, and relationships within the system.

The main problems addressed in this milestone include:

- i. Academic management operations
- ii. Assessment and submission workflows
- iii. Faculty teaching assignments
- iv. Student enrollment history
- v. AI-based evaluation processing
- vi. Automated report generation functionalities
- vii. Data consistency and scalability
- viii. Preservation of historical academic records
- ix. Proper representation of real-world university scenarios

1.3. PROJECT OBJECTIVES:

The main objective of this milestone is to design a complete and normalized ERD/EERD for the AI-Based Academic Assessment and Report Generation System that accurately represents academic workflows and relationships within a university environment. The specific objectives of this milestone are as follows:

- 1. To identify all major entities required in the system.
- 2. To define appropriate attributes for each entity.
- 3. To model supertype-subtype relationships for user management.
- 4. To identify associative entities for many-to-many relationships.
- 5. To define relationships and their cardinalities according to university workflows.
- 6. To preserve historical academic records such as enrollments and evaluations.
- 7. To represent departments, programs, batches, semesters, sessions, and sections in a structured manner.
- 8. To support repeat-course enrollment and faculty teaching assignments across programs.
- 9. To support AI-based evaluation and automated report generation.
- 10. To ensure normalization for reducing redundancy and improving consistency.
- 11. To create a professional ERD/EERD using standard Chen notation

1.4. GITHUB REPOSITORY LINK:

Please click the link for the GitHub repository. [[Link](#)]

CHAPTER 2: DETAILED DATABASE DESIGN

2.1. ENTITY:

Sr. No	Entity Name	Description
01	User	Stores common authentication and profile information for all system users including students, faculty members, and administrators.
02	Student	Stores student-specific academic information such as registration numbers, batch, program, and current academic details.
03	Faculty	Stores faculty member information including designation, employee details, and department association.
04	Department	Stores university department information and manages related programs, faculty members, and courses.
05	Program	Stores academic program details such as BS AI, BS Cyber Security, and BS IT offered under departments.
06	Batch	Stores admission batch information of students enrolled in a specific program.
07	Semester	Stores academic semester levels representing student study progression.
08	Session	Stores academic sessions or terms such as Spring 2026 and Fall 2025.
09	Section	Stores section information for a specific program and session.
10	Course	Stores course-related information including course code, title, and credit hours.
11	Assessment	Stores assessment details such as quizzes, assignments, and deadlines.
12	Submission	Stores student uploaded submissions for assessments.
13	Evaluation	Stores AI-generated evaluation results, marks, feedback, and performance reports.
14	User Phone	Stores multiple contact numbers associated with system users.
15	Activity Log	Stores user activities and system operation logs.

2.2. DATA DICTIONARY:

2.2.1. User:

Sr. No	Name	Data Type	Constraint	Description
01	User ID (Primary Key)	Integer	NOT NULL, PK	Unique identifier for the user
02	First Name	String	NOT NULL	First name of the user
03	Last Name	String	NOT NULL	Last name of the user
04	Email	String	NOT NULL, UNIQUE	Email address of the user

05	Password	String	NOT NULL	Encrypted password of the user
06	Role	String	NOT NULL	Role of the user (e.g., Admin, Faculty, Student)
07	Gender	String	NULLABLE	Gender of the user
08	Account Status	String	NOT NULL	Status of the user account (e.g., Active, Inactive)
09	Created At	DateTime	NOT NULL	Timestamp when the user account was created

2.2.2. User Phone

Sr. No	Name	Data Type	Constraint	Description
01	phone_ID (Primary Key)	String	NOT NULL, PK	The unique identifier for the data
02	Phone_no	Integer	Not Null	Phone number of the user

2.2.3. Student

Sr. No	Name	Data Type	Constraint	Description
01	Student_ID (Primary Key)	Integer	NOT NULL, PK	Unique identifier for the student
02	registration_no	String	NOT NULL, UNIQUE	Official registration number of the student
03	current_semester	Integer	NOT NULL	Current semester of the student
04	student_status	String	NOT NULL	Current status of the student (e.g., Enrolled, Graduated)

2.2.4. Faculty

Sr. No	Name	Data Type	Constraint	Description
01	Teacher_ID (Primary Key)	Integer	NOT NULL, PK	Unique identifier for the faculty member
02	Designation	String	NOT NULL	Academic designation of the faculty member

2.2.5. Program

Sr. No	Name	Data Type	Constraint	Description
01	program_ID (Primary Key)	Integer	NOT NULL, PK	Unique identifier for the academic program
02	program_name	String	NOT NULL	Name of the academic program
03	program_code	String	NOT NULL, UNIQUE	Short code for the program
04	degree_level	String	NOT NULL	Level of degree (e.g., Undergraduate, Graduate)
05	total_semester	Integer	NOT NULL	Total number of semesters in the program

2.2.6. Activity Log

Sr. No	Name	Data Type	Constraint	Description
01	Log_id (Primary Key)	Integer	NOT NULL, PK	Unique identifier for the activity log entry
02	Activity_description	Text	NOT NULL	Detailed description of the activity
03	Time	DateTime	NOT NULL	Timestamp when the activity occurred

2.2.7. Department

Sr. No	Name	Data Type	Constraint	Description
01	Department_ID (Primary Key)	Integer	NOT NULL, PK	Unique identifier for the department
02	Department Name	String	NOT NULL	Full name of the department
03	Department_Code	String	NOT NULL, UNIQUE	Short code representing the department

2.2.8. Batch

Sr. No	Name	Data Type	Constraint	Description
01	Batch_ID (Primary Key)	Integer	NOT NULL, PK	Unique identifier for the batch
02	Batch_name	String	NOT NULL	Name of the batch
03	Expected_Graduation	Date	NULLABLE	Expected graduation date for the batch

04	Batch_status	String	NOT NULL	Current status of the batch
05	start_year	Integer	NOT NULL	The year the batch started

2.2.9. Section

Sr. No	Name	Data Type	Constraint	Description
01	Section ID (Primary Key)	Integer	NOT NULL, PK	ID or identifier for the section
02	capacity	Integer	NOT NULL	Maximum number of students allowed in the section
03	Section Name	String	Not Null	Name or identifier for the section

2.2.10. Session

Sr. No	Name	Data Type	Constraint	Description
01	session_ID (Primary Key)	Integer	NOT NULL, PK	Unique identifier for the session
02	session_name	String	NOT NULL	Name of the session
03	Academic_year	String	NOT NULL	Academic year of the session
04	start_date	Date	NOT NULL	Start date of the session
05	end_date	Date	NOT NULL	End date of the session

2.2.11. Teacher Assignment

Sr. No	Name	Data Type	Constraint	Description
01	Assignment_ID (Primary Key)	Integer	NOT NULL, PK	Unique identifier for the teacher assignment
02	Assigned Date	Date	NOT NULL	Date when the assignment was made

2.2.12. Course

Sr. No	Name	Data Type	Constraint	Description
01	Course_ID (Primary Key)	Integer	NOT NULL, PK	Unique identifier for the course
02	Course_Code	String	NOT NULL, UNIQUE	Official code for the course

03	Course_Title	String	NOT NULL	Full title of the course
04	Credit_Hours	Integer	NOT NULL	Number of credit hours for the course

2.2.13. Enrollment

Sr. No	Name	Data Type	Constraint	Description
01	Enrollment_ID (Primary Key)	Integer	NOT NULL, PK	Unique identifier for the enrollment record
02	is_repeat	Boolean	NOT NULL	Indicates if the course is being repeated
03	Enrollment_Status	String	NOT NULL	Status of the enrollment (e.g., Pending, Approved)
04	Enrollment_Start_Date	Date	NOT NULL	Date when enrollment started
05	Enrollment_End_Date	Date	NULLABLE	Date when enrollment ended

2.2.14. Semester

Sr. No	Name	Data Type	Constraint	Description
01	Semester_ID (Primary Key)	Integer	NOT NULL, PK	Unique identifier for the semester
02	Semester_no	Integer	NOT NULL	The number of the semester in the sequence
03	Academic_year	String	NOT NULL	The academic year of the semester
04	start_date	Date	NOT NULL	Start date of the semester
05	end_date	Date	NOT NULL	End date of the semester

2.2.15. Assessment

Sr. No	Name	Data Type	Constraint	Description
01	Assessment_ID (Primary Key)	Integer	NOT NULL, PK	Unique identifier for the assessment
02	Title	String	NOT NULL	Title of the assessment
03	Description	Text	NULLABLE	Detailed description of the assessment

04	Assessment_Type	String	NOT NULL	Type of assessment (e.g., Quiz, Assignment, Exam)
05	Total Marks	Integer	NOT NULL	Maximum marks possible for the assessment
06	Created_At	DateTime	NOT NULL	Timestamp when the assessment was created
07	Deadline	DateTime	NOT NULL	Submission deadline for the assessment
08	class_report_path	String	NULLABLE	File path to the generated class report
09	class_report_generated_at	DateTime	NULLABLE	Timestamp when the class report was generated
10	class_report_statuses	String	NULLABLE	Current status of the class report generation

2.2.16. Registered Course

Sr. No	Name	Data Type	Constraint	Description
01	Registered_Course_ID (Primary Key)	Integer	NOT NULL, PK	Unique identifier for the registered course record

2.2.17. Evaluation

Sr. No	Name	Data Type	Constraint	Description
01	Evaluation_ID (Primary Key)	Integer	NOT NULL, PK	Unique identifier for the evaluation
02	Obtained_Marks	Decimal	NOT NULL	Marks obtained by the student
03	Percentage	Decimal	NOT NULL	Percentage score calculated from marks
04	Evaluated_At	DateTime	NOT NULL	Timestamp when the evaluation was performed
05	Evaluation_Status	String	NOT NULL	Status of the evaluation (e.g., Graded, Re-evaluating)
06	student_report_path	String	NULLABLE	File path to the student's individual report

07	student_report_generated_at	DateTime	NULLABLE	Timestamp when the student report was generated
08	student_report_status	String	NULLABLE	Status of the student report generation

2.2.18. Submission

Sr. No	Name	Data Type	Constraint	Description
01	Submission_ID (Primary Key)	Integer	NOT NULL, PK	Unique identifier for the submission
02	File_Name	String	NOT NULL	Name of the submitted file
03	File_Path	String	NOT NULL	Storage path of the submitted file
04	Upload_Time	DateTime	NOT NULL	Timestamp when the file was uploaded
05	Submission Status	String	NOT NULL	Status of the submission (e.g., Submitted, Late)

2.3. BUSINESS RULES:

- Each user in the system must have a unique user account for authentication and system access.
- A user can exist either as a student or a faculty member within the system.
- A user can have multiple contact numbers stored in the system.
- Each student must belong to one specific program throughout the degree duration.
- Each student must belong to one admission batch of a program.
- A department can offer multiple academic programs, but each program belongs to only one department.
- A department can offer multiple courses, but each course belongs to only one department.
- A program can contain multiple sections, but each section belongs to one specific program.
- Each section is conducted within one academic session, while a session can contain multiple sections.
- A faculty member belongs to one department but can teach multiple courses and sections.
- A faculty member can have multiple teaching assignments in different semesters and sections.
- Each teaching assignment is associated with one faculty member, one course, one section, and one semester.

- A course can be taught in multiple sections and semesters through different teaching assignments.
- A student can have multiple enrollment records during different semesters and sessions.
- Each enrollment record belongs to one student and stores the student's semester, section, and session information.
- A single enrollment can contain multiple registered courses through Course Enrollment records.
- Each Course Enrollment record links one enrollment with one teaching assignment.
- A student can repeat a failed course in another semester or section through a new enrollment record.

2.4. ENTITY RELATIONSHIP DIAGRAM:

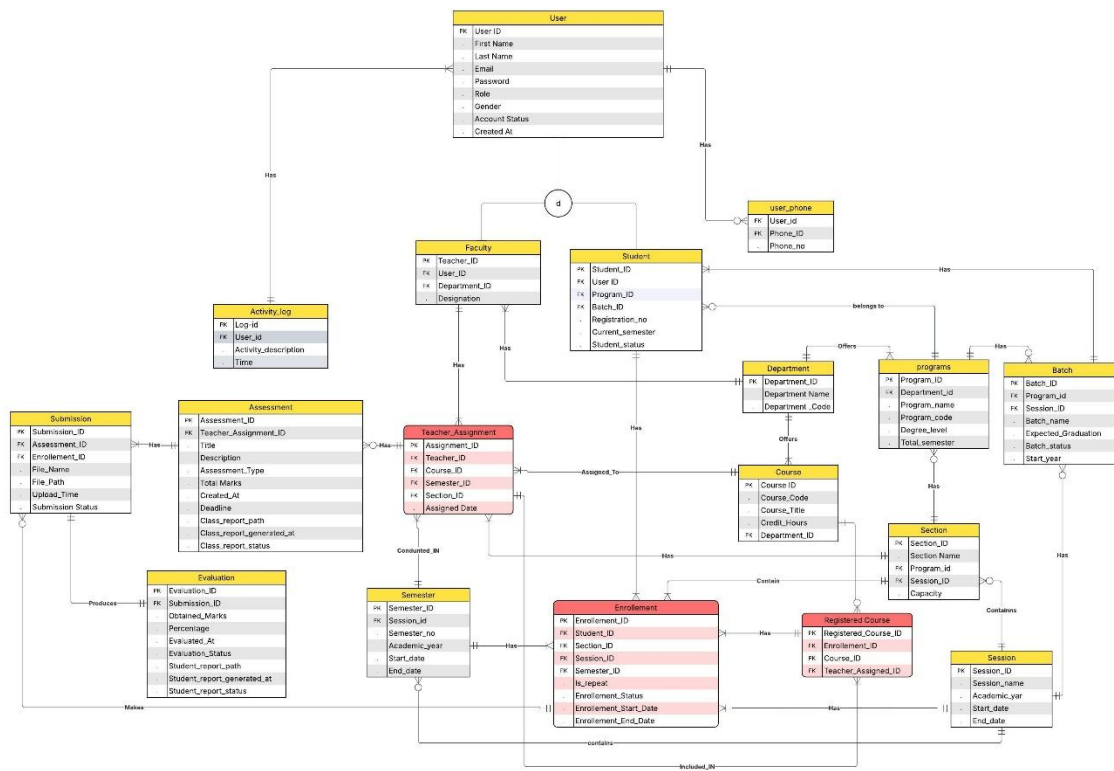


Figure 1: Enhanced Entity Relationship Diagram

References

- [1] Lucid Software Inc., “What is an Entity Relationship Diagram (ERD)?,” *Lucidchart*, 2024. [Online]. Available: <https://www.lucidchart.com/pages/er-diagrams>. [Accessed: 18-May-2026].
- [2] Oracle Corporation, “Database Design Basics,” *Oracle Documentation*, 2024. [Online]. Available: <https://docs.oracle.com/en/database/>. [Accessed: 18-May-2026].
- [3] Creately, “Entity Relationship Diagram Symbols and Notation,” *Creately*, 2024. [Online]. Available: <https://creately.com/guides/er-diagram-symbols/>. [Accessed: 18-May-2026].

Chapter 4 APPENDIX: AI USAGE AND PROMPTS

4.1 PURPOSE OF AI USAGE

Artificial Intelligence (AI) tools were used during the design and documentation phase of the Entity Relationship Diagram (ERD/EERD) milestone for the proposed AI-Based Academic Assessment and Report Generation System. The AI tool was mainly used for ERD consultation, relationship validation, normalization guidance, business rule formulation, and report writing assistance.

The final ERD structure, assumptions, and academic workflow decisions were finalized according to the project requirements and university scenarios.

4.2 AI TOOL USED

Tool Name	Purpose
ChatGPT (OpenAI)	ERD consultation, relationship validation, normalization guidance, and report writing assistance

4.3 PROMPTS USED DURING DEVELOPMENT

Sr. No	Prompt Used	Purpose
1	"Now tell me the final entities and attributes."	To finalize the entities and attributes of the ERD
2	"The enrollment is associative and how many entities links to it?"	To understand associative entity relationships
3	"Now from this enhancement give me other relation also carefully understanding the system."	To refine relationship structures
4	"What is the cardinality of the relation with program and session?"	To validate cardinalities and participation constraints
5	"Now analyze this ERD and tell me. Analyze in all scenarios and assumptions."	To validate the final ERD design
6	"Now you have my ERD but some relation names missing in this so tell me every single relation name."	To finalize relationship naming
7	"Write 15–25 strong rules according to your system."	Generating business rules
8	"Add assumptions very carefully because this will defend my ERD."	To document system assumptions
9	"Explain supertype/subtype according to my ERD."	To validate inheritance modeling
10	"Make the proper flow of this report headings."	To structure the milestone report
11	"Mention all the entities in table form."	To prepare report documentation
12	"Mention all the attributes in table form for each entity."	To document entity attributes professionally

4.4 AI USAGE SUMMARY

The AI tool assisted in identifying and refining entities, designing associative entities, validating cardinalities and relationships, creating business rules and assumptions, improving normalization decisions, and structuring the ERD report professionally.

4.5 LINK OF PROMPTS

You can access the full chat history and prompt interactions at the following link: [[CHAT LINK HERE](#)]